

INTERNSHIP PROJECT REPORT

Tata GenAI Powered Data Analytics Virtual Experience

Submitted by:

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B.Tech CSE (Batch 2024–2028)

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1. Candidate's Declaration

I hereby declare that the internship report entitled "Tata GenAI Powered Data Analytics Virtual Experience" is an authentic record of my own work carried out as a virtual intern via the Forage platform in collaboration with Tata Insights and Quants (Tata iQ). This report is submitted in partial fulfillment of the requirements for the B.Tech Computer Science and Engineering program at IILM University, Greater Noida. The matter embodied in this report has not been submitted by me for the award of any other degree or diploma.

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Date: September 20, 2026

2. Acknowledgement

I would like to express my sincere gratitude to Tata Insights and Quants and the Forage platform for providing this immersive virtual experience. The simulated environment allowed me to apply theoretical data analytics and machine learning concepts to real-world financial challenges.

I also extend my deep appreciation to the faculty and administration of IILM University, Greater Noida, for their continuous support, academic guidance, and for encouraging practical, industry-aligned learning during the 5th semester of my B.Tech CSE program.

3. Internship Completion Certificate

(Attach your downloaded Tata_GenAI_Internship_Certificate.pdf here in your final Word/Doc file)

4. Project Description

4.1 Introduction

The integration of Artificial Intelligence and advanced data analytics is transforming the financial services sector. This internship project focused on applying Generative AI principles, machine learning, and data storytelling to mitigate credit risk for a simulated financial institution, Geldium. The project encompassed the end-to-end data lifecycle: from raw data exploration to structural model planning, and finally, executive strategy formulation.

4.2 Organization Profile

Tata Insights and Quants (Tata iQ): A premier analytics and AI center of excellence within the Tata Group. Tata iQ leverages deep domain expertise, advanced analytics, and data engineering to drive strategic decision-making and operational efficiency across various global industries. This virtual experience was hosted in partnership with Forage.

4.3 Problem Statement

Geldium, a financial services provider, faced challenges in proactively identifying borrowers at high risk of credit delinquency. Traditional reactive collection methods were resulting in high operational costs and increased Non-Performing Assets (NPAs). The objective was to utilize historical financial data to predict default probabilities and design an automated, AI-driven intervention strategy.

4.4 Project Objectives

Perform Exploratory Data Analysis (EDA) on borrower datasets to identify key default risk indicators.

Design a robust machine learning architecture capable of predicting delinquency probability.

Formulate a segmented, AI-driven collections strategy based on model outputs.

Establish responsible AI guardrails ensuring fairness, explainability, and regulatory compliance.

4.5 Methodology

The project followed a structured data science workflow:

Data Preprocessing & EDA: Conducted univariate and bivariate analysis on borrower data. Identified bimodal income distributions and extreme Debt-to-Income (DTI) outliers. Addressed missing values using median imputation grouped by employment type to preserve data integrity.

Feature Engineering: Selected critical predictors including Revolving Utilization Ratio, DTI, and Recent Credit Inquiries.

Model Selection: Proposed an ensemble tree-based algorithm (Random Forest / XGBoost) for its robustness against outliers and non-linear data relationships.

Evaluation Metrics: Prioritized Recall (Sensitivity) to capture maximum true defaulters, alongside F1-Score and ROC-AUC for balanced classification performance.

4.6 Collections Strategy

Based on the predictive model, a segmented customer intervention strategy was designed:

Low-Risk Tier (<15% Probability): Automated, soft-touch digital reminders (SMS/Email) using GenAI-drafted copy.

Medium-Risk Tier (15% - 50% Probability): Proactive engagement via conversational AI or human agents offering flexible restructuring options before severe arrears.

High-Risk Tier (>50% Probability): Immediate escalation to specialized human collections agents for settlement negotiation and loss mitigation.

4.7 Responsible AI Guardrails

To ensure ethical deployment of the predictive system, the following guardrails were established:

Fairness: Routine statistical auditing to achieve demographic parity and eliminate systemic bias against protected classes.

Explainability: Implementation of SHAP (SHapley Additive exPlanations) values to provide transparency into how the model calculates individual risk scores.

Compliance: Strict adherence to Fair Debt Collection Practices and ensuring a mandatory "Human-in-the-Loop" override for disputed automated decisions.

4.8 Expected Outcomes

The proposed AI system is projected to deliver significant business value:

Quantitative: Reduction in overall NPAs, 20% decrease in operational cost-to-collect, and a 15% increase in early-stage recovery rates.

Qualitative: Improved customer experience through personalized, proactive financial assistance and the preservation of long-term brand loyalty during periods of financial hardship.

5. Bibliography/References

Tata Insights and Quants Virtual Experience Program. Forage. <https://www.theforage.com>

Tata Group Official Portal. <https://www.tata.com>

Documentation on Ensemble Machine Learning Methods (Random Forest, XGBoost).

Guidelines for Responsible AI and Fair Debt Collection Practices.